BAHIR-DAR UNIVERSITY
BAHIR-DAR INSTITUTE OF TECHNOLOGY(BiT)
Faculty of Computing
Department of SOFTWARE Engineering
OSSP - Individual PROJECT

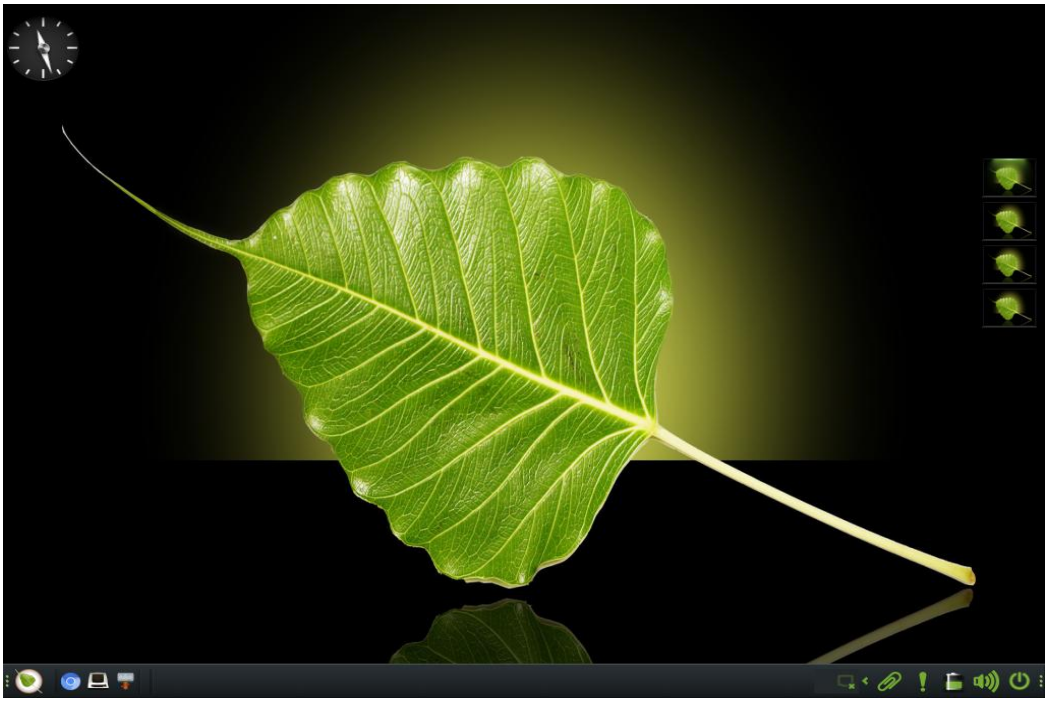
Title : Operating Systems (Linux Installation)
Operating System : Bodhi Linux 7.0.0 (Standard Edition)
Virtualization Software : Oracle VM VirtualBox 7.x

Name	ID
Yeabisira Challachew	BDU1605027

Submitted to: Lec.Wendimu Baye
Submission Date: 20/08/2018E.C

WELCOME TO

Bodhi-LiNUX Operating System



A. Introduction

Background

Bodhi Linux is a lightweight, Ubuntu-based Linux distribution that features the Moksha Desktop Environment. The word "Bodhi" is rooted in the Sanskrit term for "enlightenment," which reflects the project's philosophy of providing a minimalist yet functional user experience. Unlike many mainstream operating systems that come bloated with pre-installed software, Bodhi Linux follows a "Minimalist" approach, providing only the essential tools (a terminal, a web browser, and a text editor) to keep the system fast and responsive.

The Moksha Desktop, a fork of Enlightenment 17, allows for high levels of customization while maintaining an extremely low hardware footprint. This makes Bodhi Linux an ideal candidate for both modern hardware and older machines that struggle with resource-heavy operating systems like Windows or standard GNOME-based Linux distros.

Motivation

The selection of Bodhi Linux for this project is driven by several academic and practical motivations:

- **Resource Optimization:** To demonstrate how an operating system can achieve high performance on limited hardware (low RAM and CPU) through efficient system design.

- **Minimalist System Architecture:** To study the "minimalist" philosophy of OS design, where the user has total control over which packages and dependencies are added to the kernel and desktop environment.

- **Stability and Support:** Since Bodhi is built on Ubuntu LTS (Long Term Support), it provides a stable environment for testing system calls and system programming without the risk of frequent system crashes.

- **Educational Exploration of Virtualization:** To practically apply the concepts of virtualization by installing a niche OS within a controlled virtual environment (VirtualBox/VMware), allowing for a deep dive into filesystem support and kernel-level configurations.

B. Objectives

The primary goal of this project is to gain hands-on experience with the installation, configuration, and management of the Bodhi Linux operating system within a virtual environment.

The specific objectives are as follows:

- To successfully set up and configure a virtual machine using tools like VMware Workstation or Oracle VM VirtualBox to host a guest operating system (**Virtualization Mastery).**
- To perform a complete installation of Bodhi Linux, ensuring that user accounts are created using my full name as per the project requirements (**System Installation & Configuration).**
- To identify, document, and resolve any hardware or software issues encountered during the installation process (**Technical Problem Solving).**
- To evaluate and implement the most suitable filesystem (such as ext4) for Bodhi Linux and understand the advantages and disadvantages of various filesystem types (**Filesystem Analysis).**
- To research and describe the significance of UNIX standardization, including ISO C, IEEE POSIX, and the evolution of various UNIX implementations (**UNIX Standards Understanding).**
- To observe system-level issues and develop automated solutions using system programming techniques, specifically focusing on shell scripting or batch scripts (**System Programming Application).**
- To explore the behavioral models of viruses and anti-virus construction to enhance understanding of file safety and system protection (**Security Awareness).**

C. Requirements

To ensure a smooth installation of Bodhi Linux within a virtual environment, the following hardware specifications and software tools are necessary:

i） Hardware Requirements (Virtual)

- ✓ **Processor/CPU**: Min: 1.0 GHz, Normal : dual-core (2) processor.
- ✓ **Memory (RAM)**: A min 768 MB , 2 GB is recommended for better performance.
- ✓ **Storage:** 20 GB of free disk space for the virtual hard drive (VHD).
- ✓ **Graphics:** A VGA-capable graphics card capable of 800x600 resolution.
- ✓ **Internet Connection:** Required for downloading the ISO file and system updates.

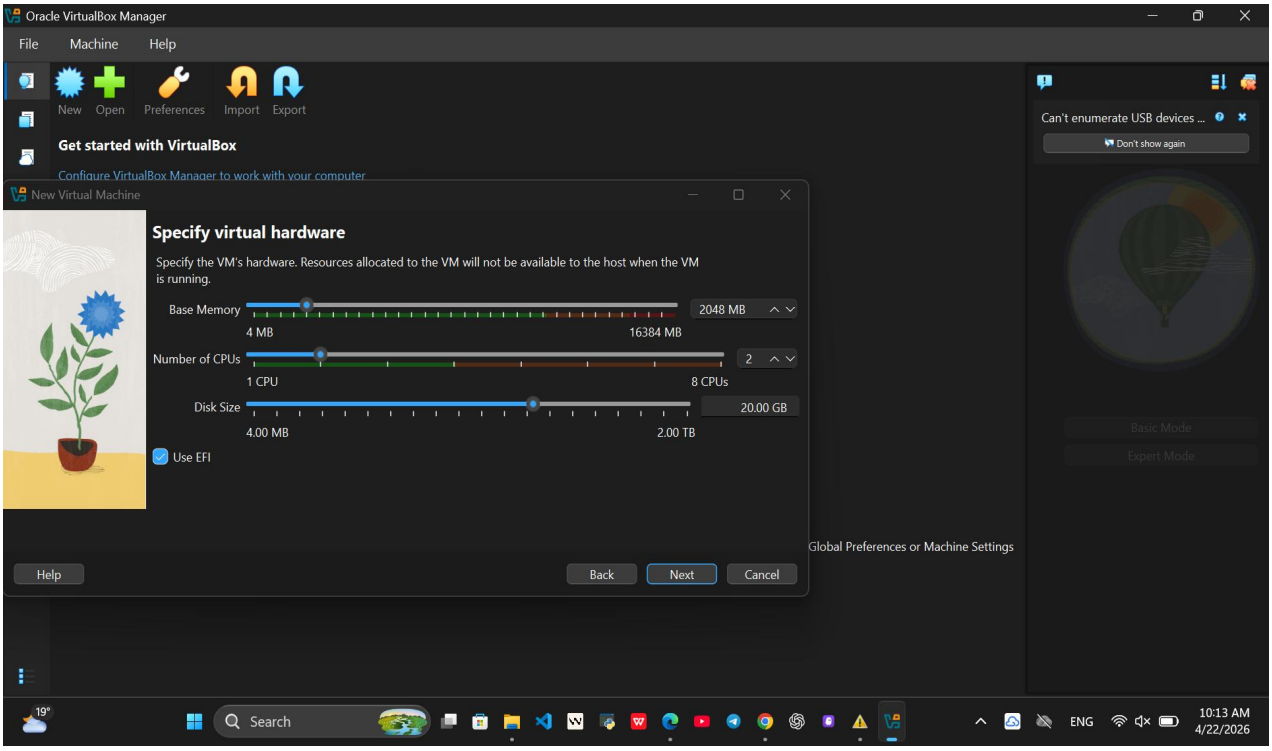


fig1 Allocating of Hardware Requirements

ii) Software Requirements

- ✓ **Host Operating System**: **Windows 10/11,** macOS, or a Linux distribution.
- ✓ **Virtualization Tool**: Oracle VM VirtualBox 7.0 (or VMware Workstation).
- ✓ **Guest Operating System ISO**: Bodhi Linux 7.0.0 ISO (Standard or HWE release).
- ✓ **Documentation Tools:** Snipping Tool (or equivalent) for capturing installation images and a word processor for the final report.
- ✓ **Programming Environment**: Built-in Linux Terminal for shell scripting and a text editor (like Leafpad or Vim) for system programming tasks.

Note: I use the **.ext4 file support system.**

D. Installation Steps

Phase 1: Environment Setup

Step 1: Accessing VirtualBox Official Website

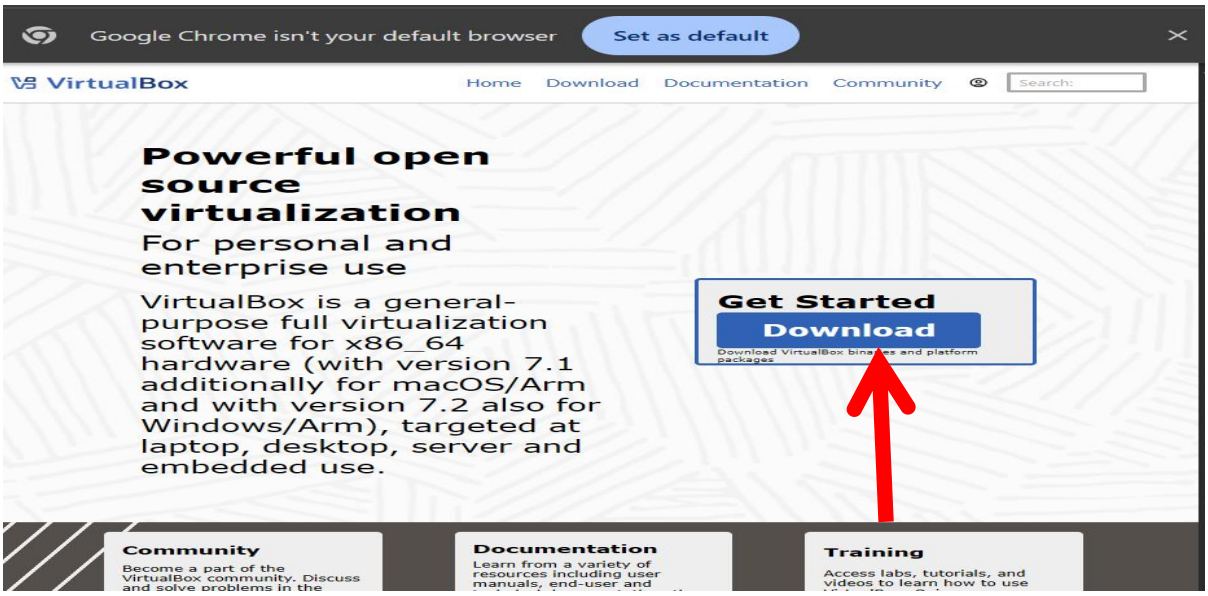


Fig1.1 Navigating to the official VirtualBox website to download the latest version of the hypervisor for Windows 11.

Step2: Choosing Windows Hosts Platform

Fig 1.2 selecting the specific installer for window hosts

Step 3: Navigating to Bodhi Linux Homepage

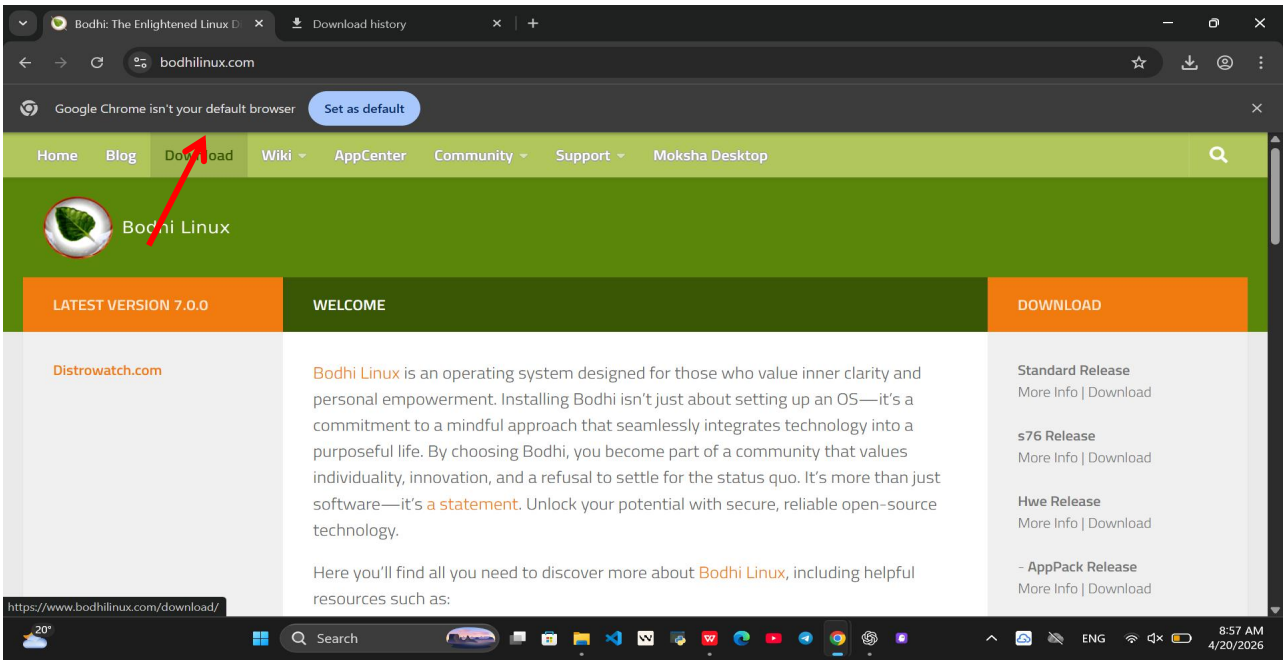


Fig 1.3 Accessing the official website of Bodhi Linux

Step 4: Choosing the Standard Edition

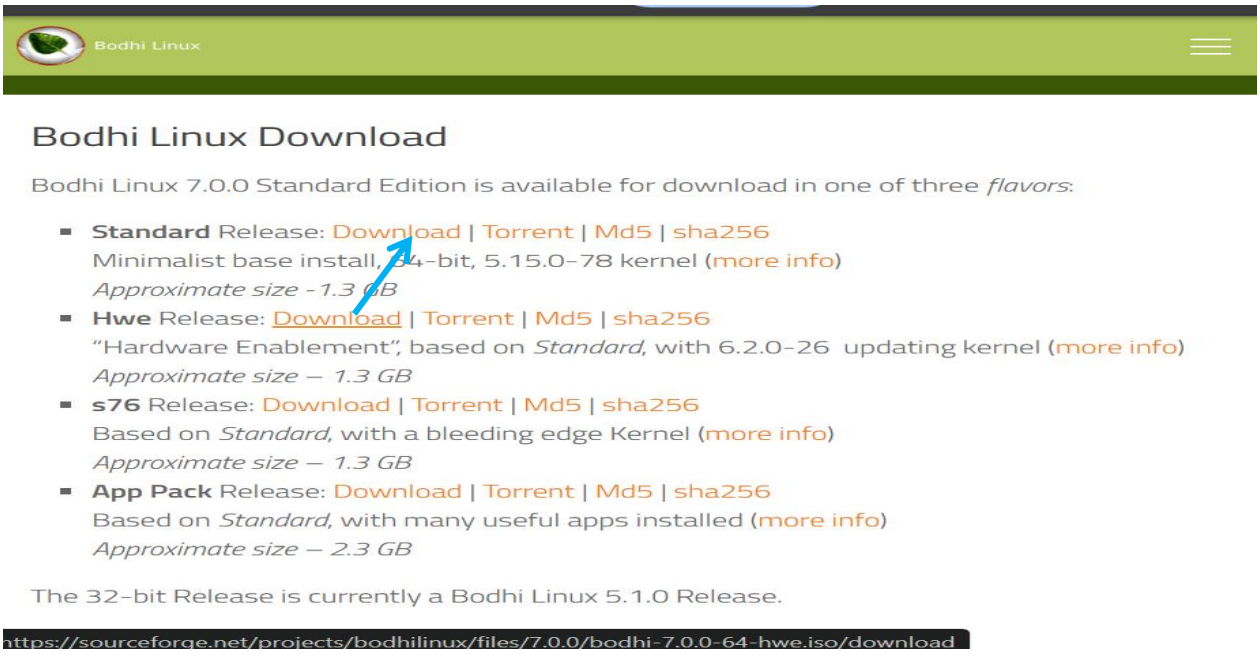


Fig 1.4 Selecting the **Standard Release (7.0.0)**

Step 5: Downloading Bodhi Linux ISO

Fig 1.5 downloading Bodhi Linux ISO

Phase 2: Virtual Machine Configuration (VirtualBox)

Step 1: Creating a New Virtual Machine

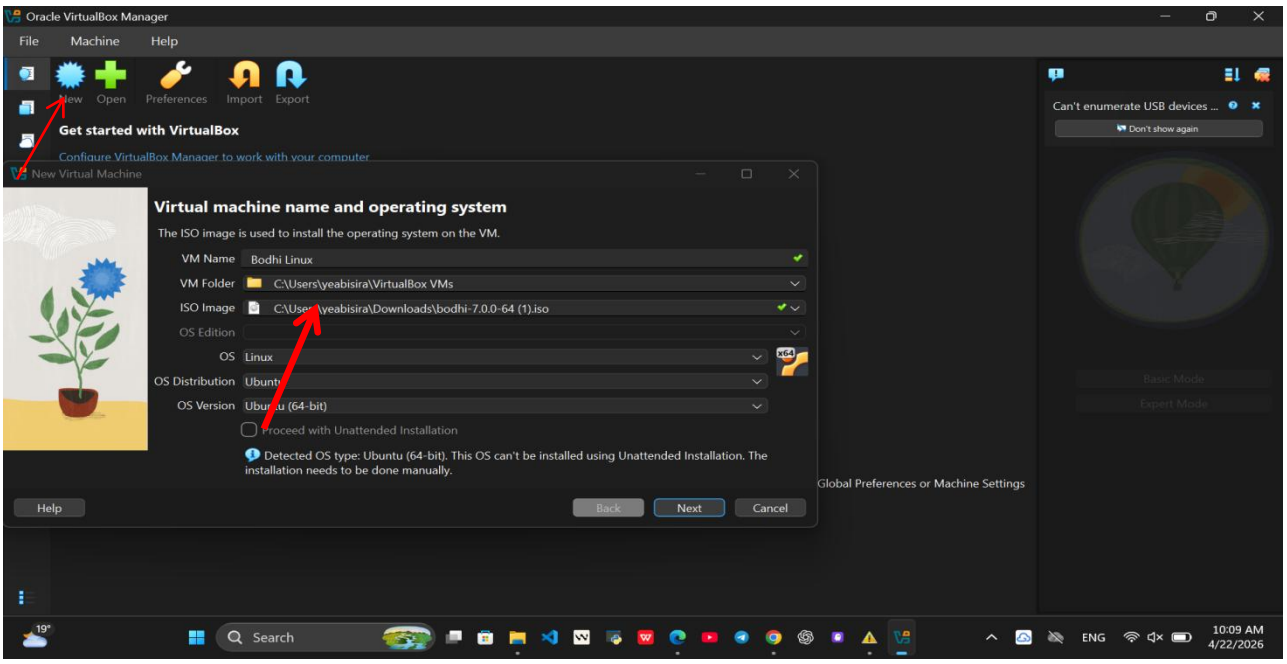


Fig 2.1 Initiating the VM creation process by naming it "Bodhi Linux"

Step 2 : Hardware Resource Allocation

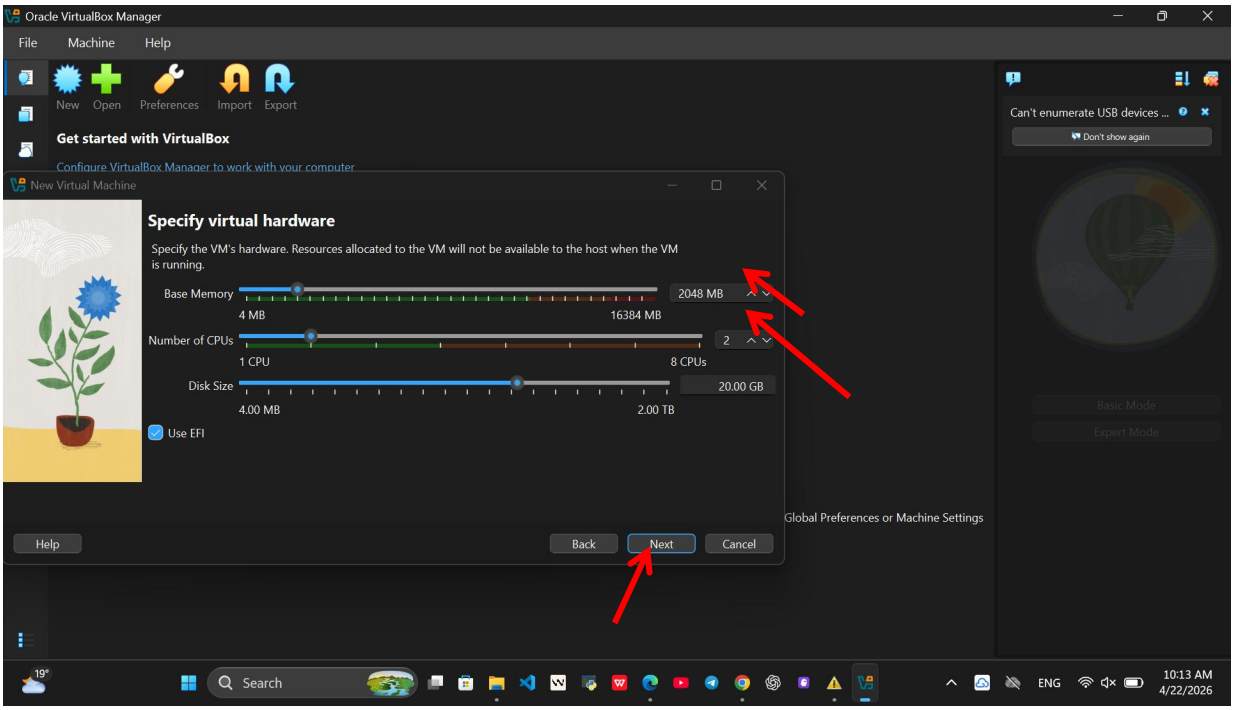


Fig 2.2 allocating RAM & CPU

Step 3: VM Configuration Summary

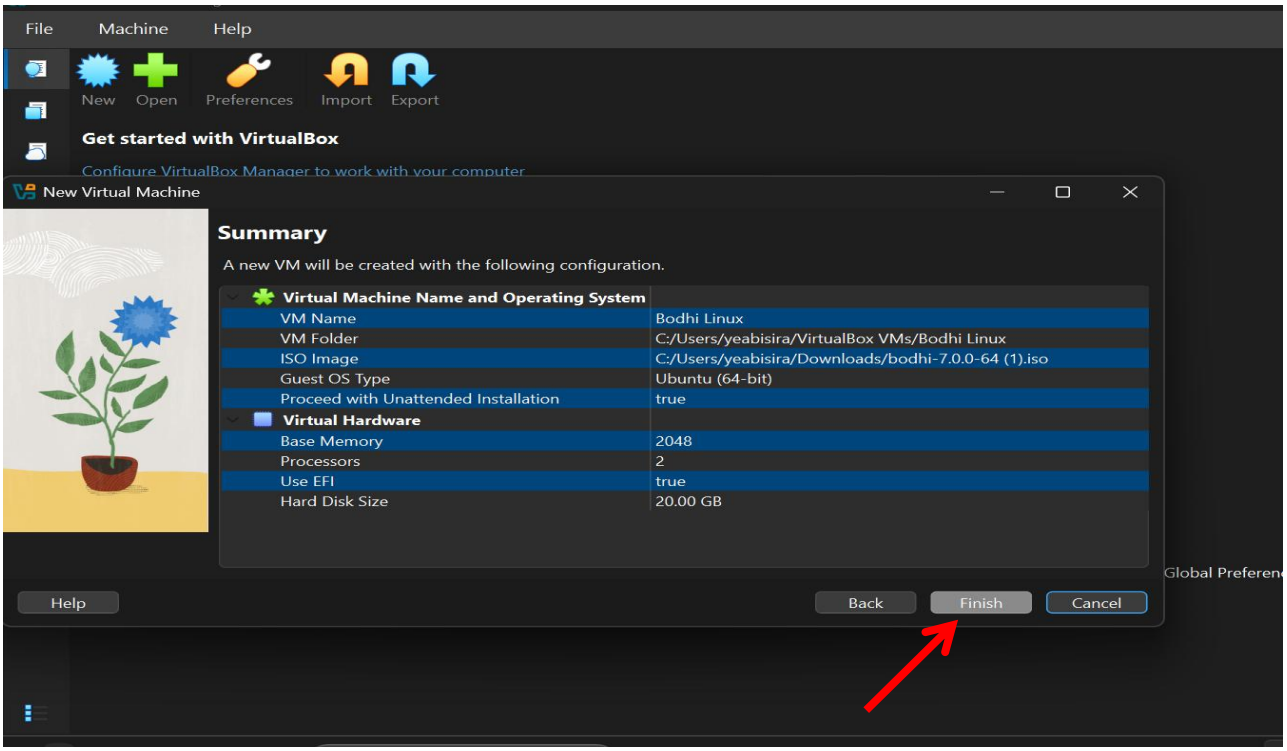


Fig 2.3 Reviewing the finalized hardware and system settings

Step 4: Virtual Machine Ready for Booting

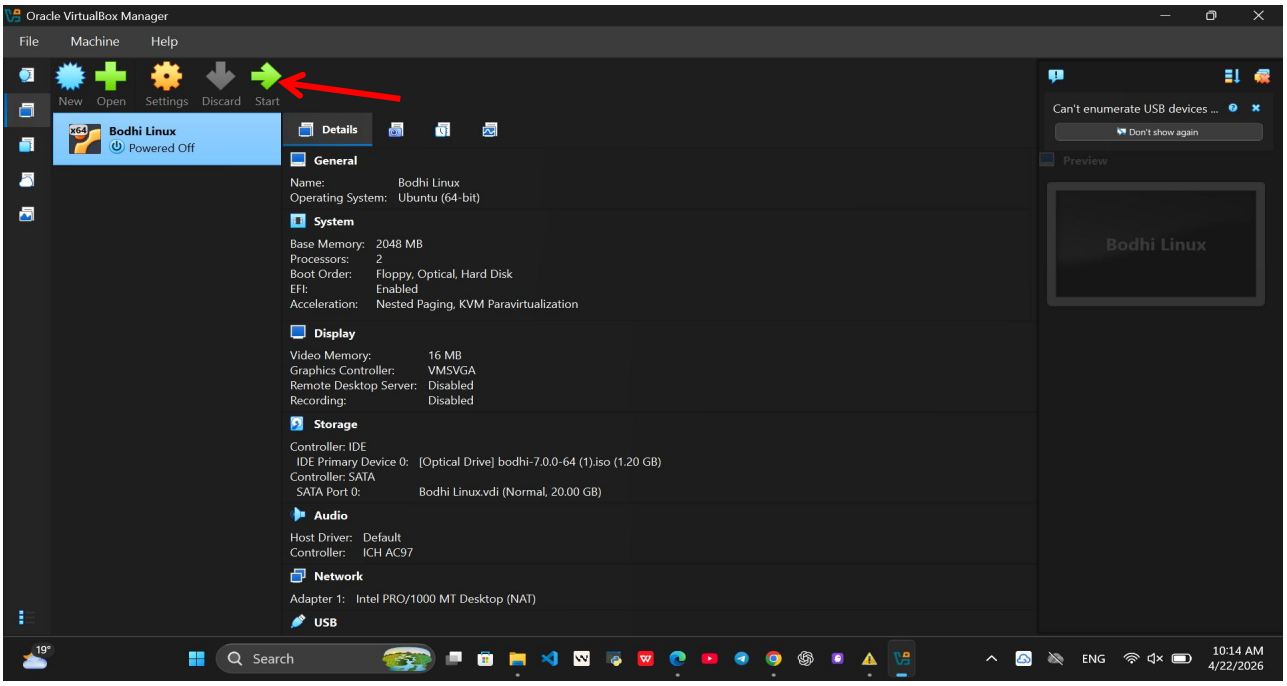


Fig 2.4 virtual machine ready to powered on for 1st time

Phase 3: Operating System Installation (Bodhi Linux)

Step 1 : Initial Boot Menu

Fig 3.1 Booting the ISO into the live environment to begin the installation process

Step 2 : Language Selection

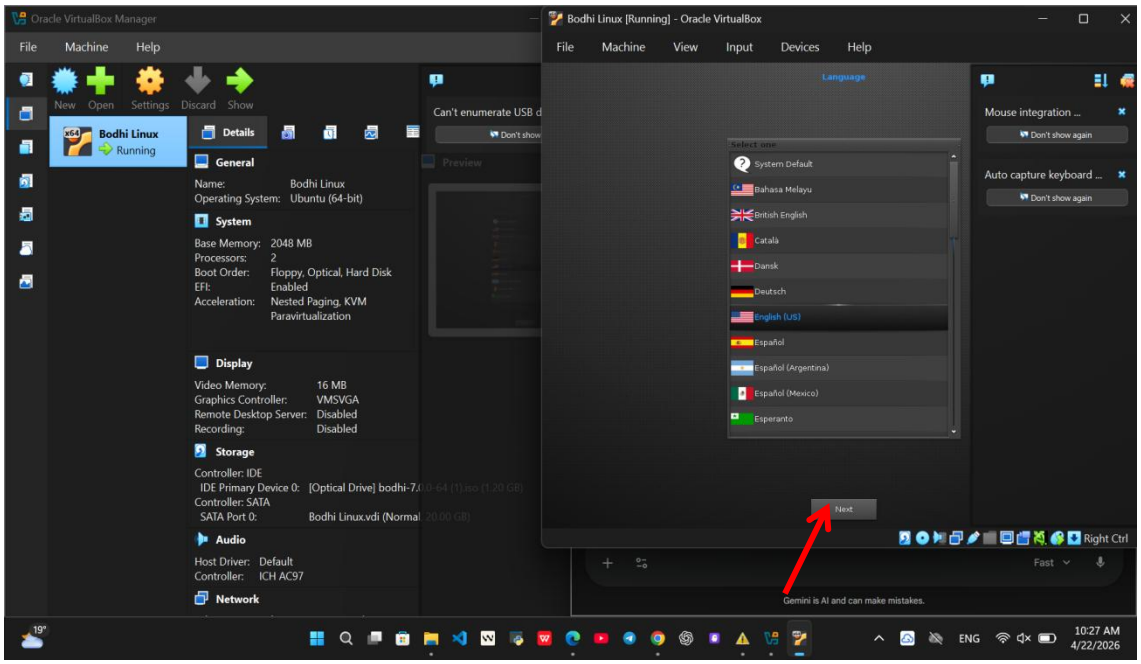


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Fig 3.2 Bodhi Linux language selection 1st page

Step 3 : Live Desktop Environment

Fig 3.3 Bodhi Linux installation home page

Step 4: Language Selection

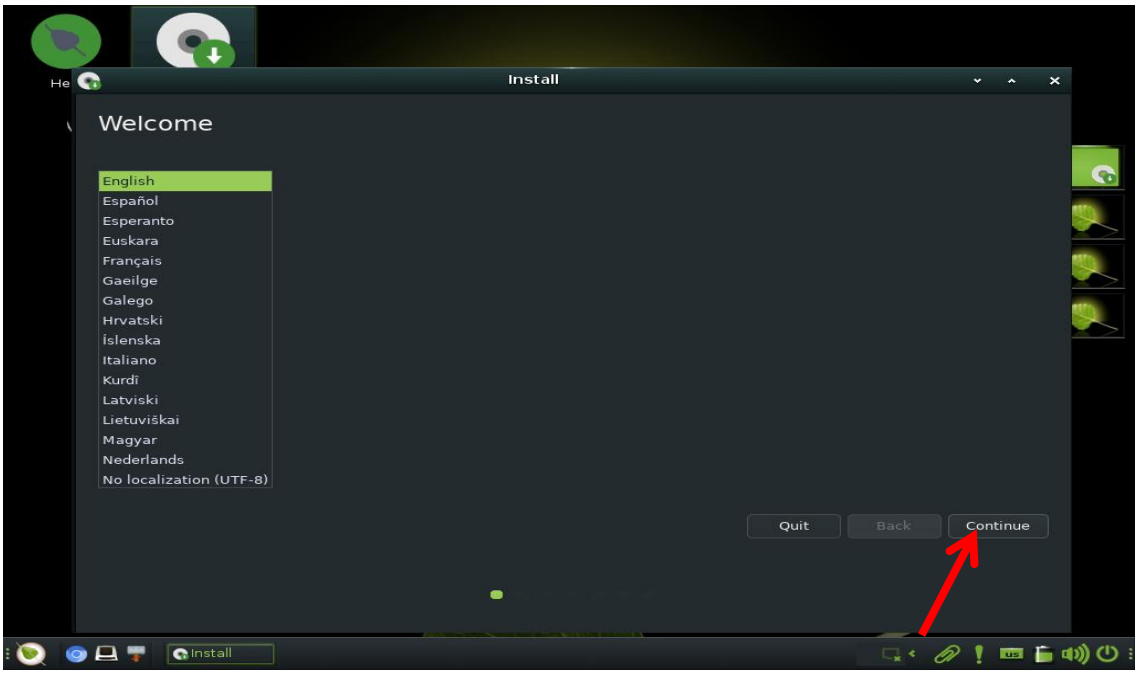

Fig 3.4 choosing language

Step 5: Keyboard Layout language selection

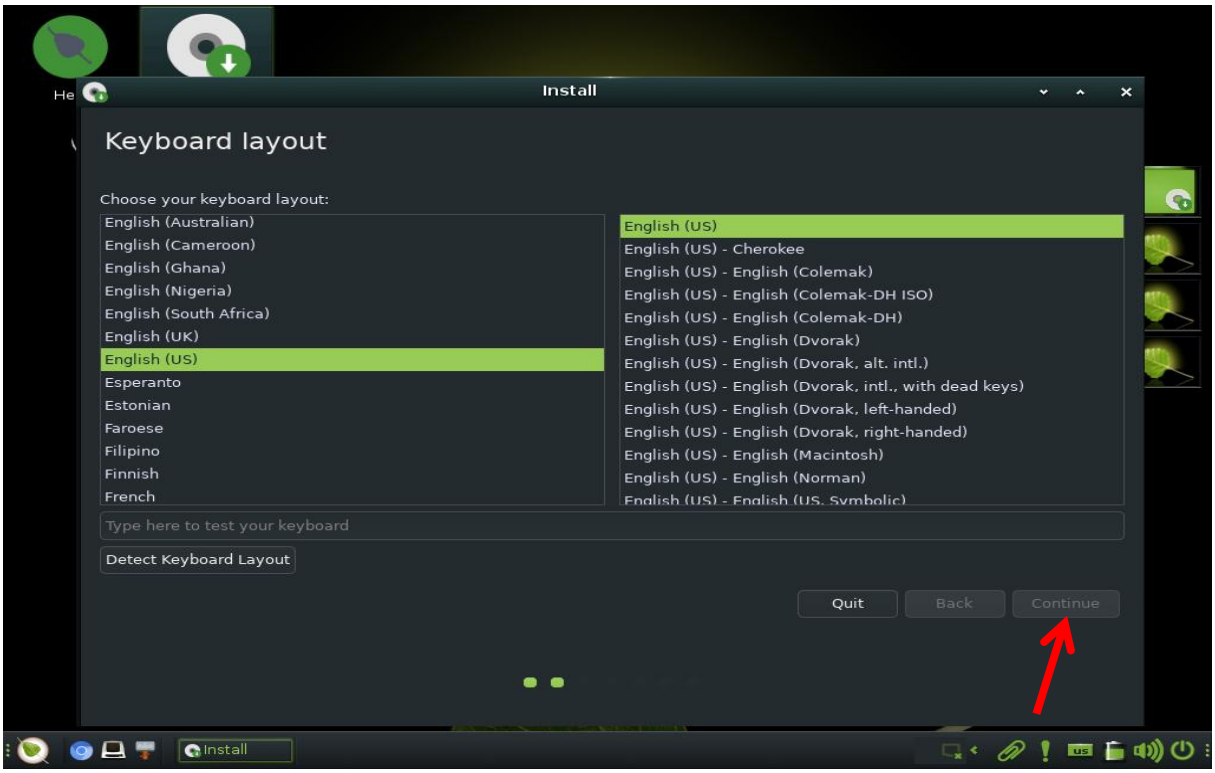

Fig 3.5 selecting keyboard layout to install Bodhi Linux

Step 6: Updates and other software check

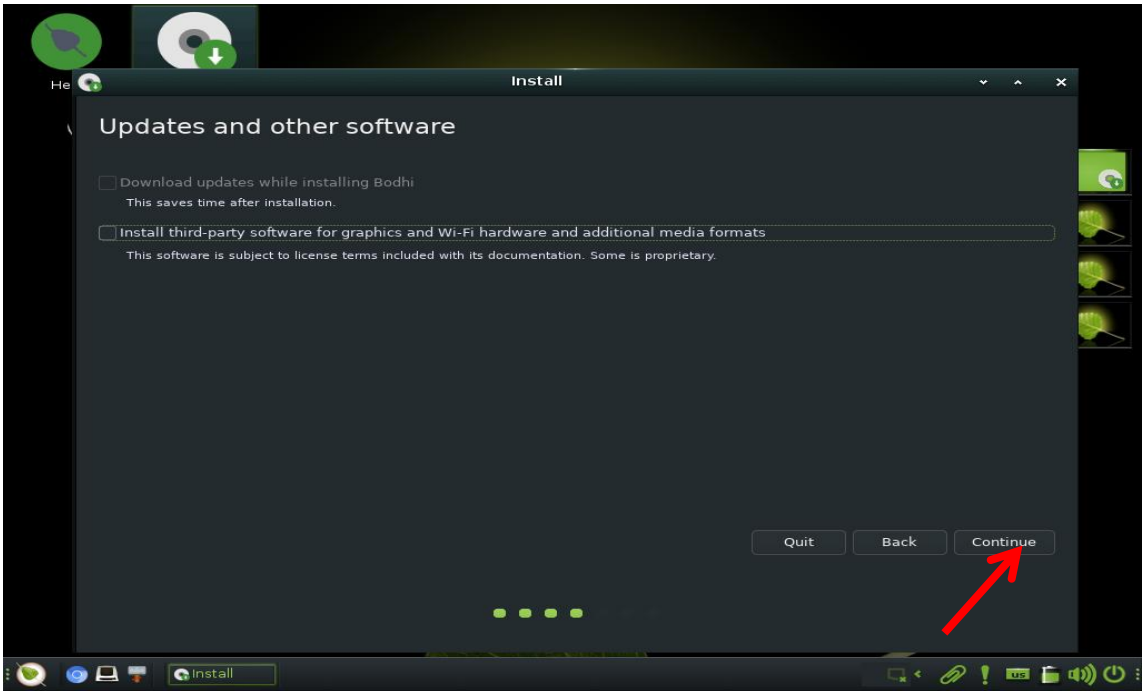

Fig 3.6 Updates and other softwares

Step 7: Installation Type selection

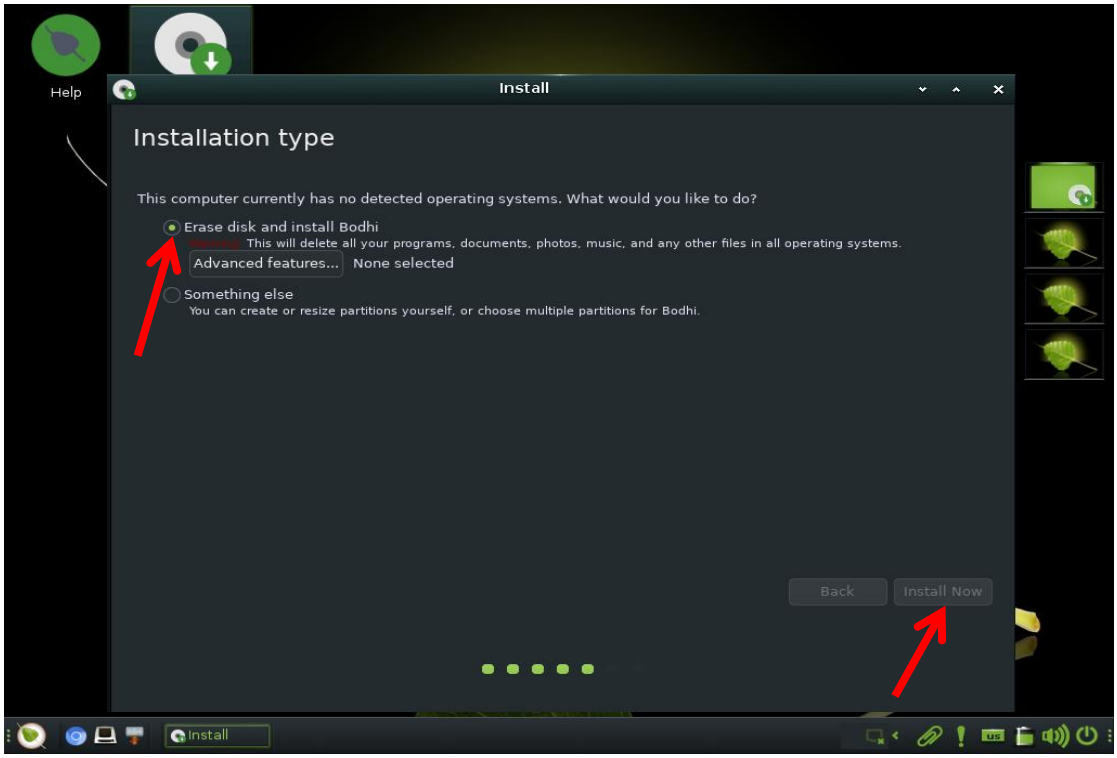

Fig 3.7 Appropriate Installation type

Step 8: Dialog Box in installation type

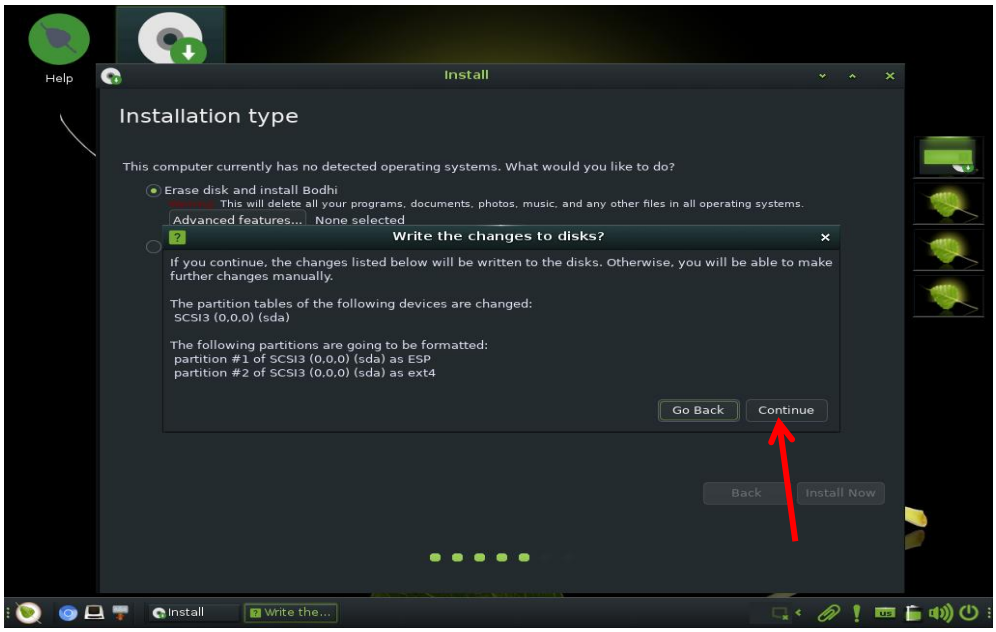

Fig 3.8 Description box in installation type

Step 9: Region and Time Zone Selection

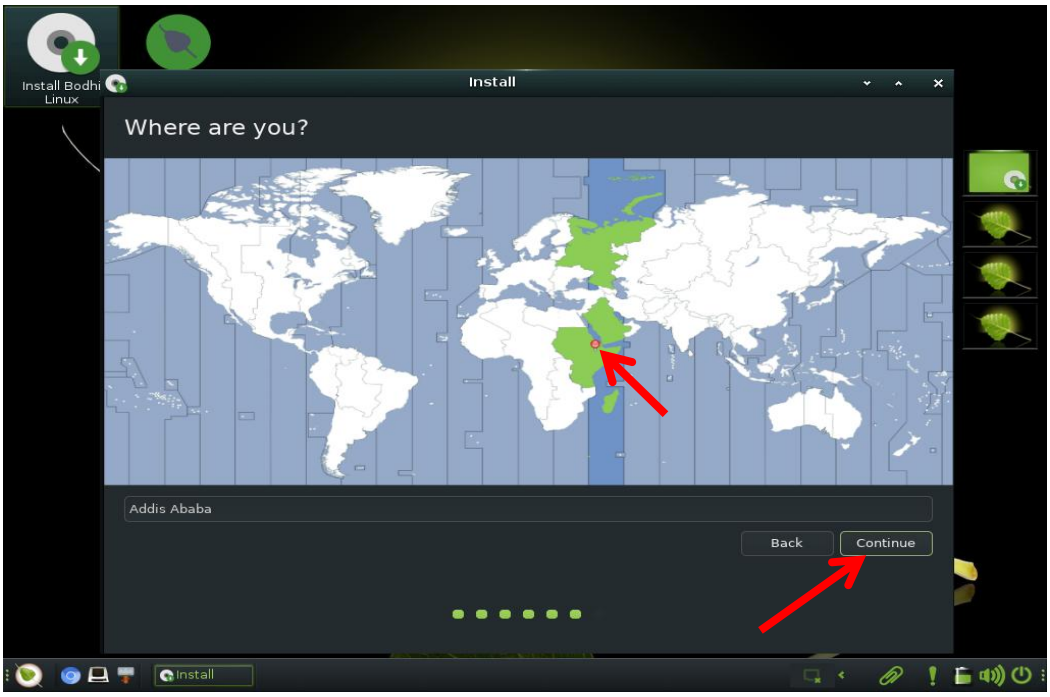

Fig 3.9 Region and Time zone selection

Step 10: User Account Creation

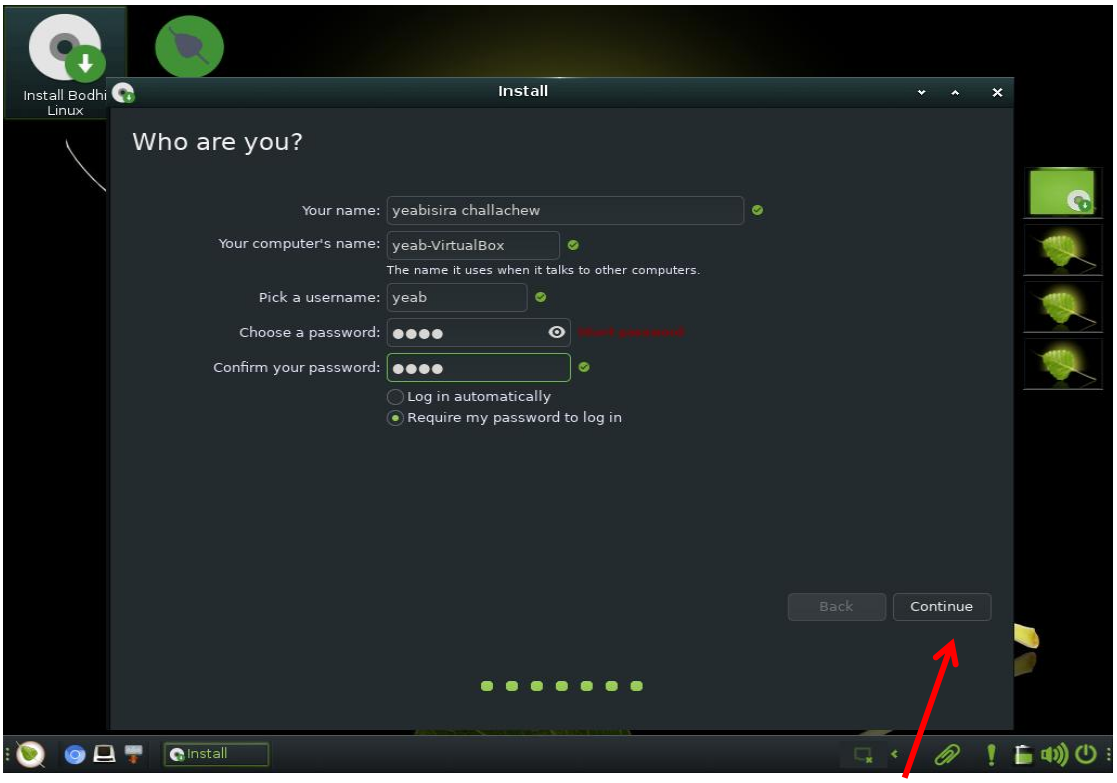

Fig 3.10 User Account Creation

Step 11: System File Copying Progress Bar

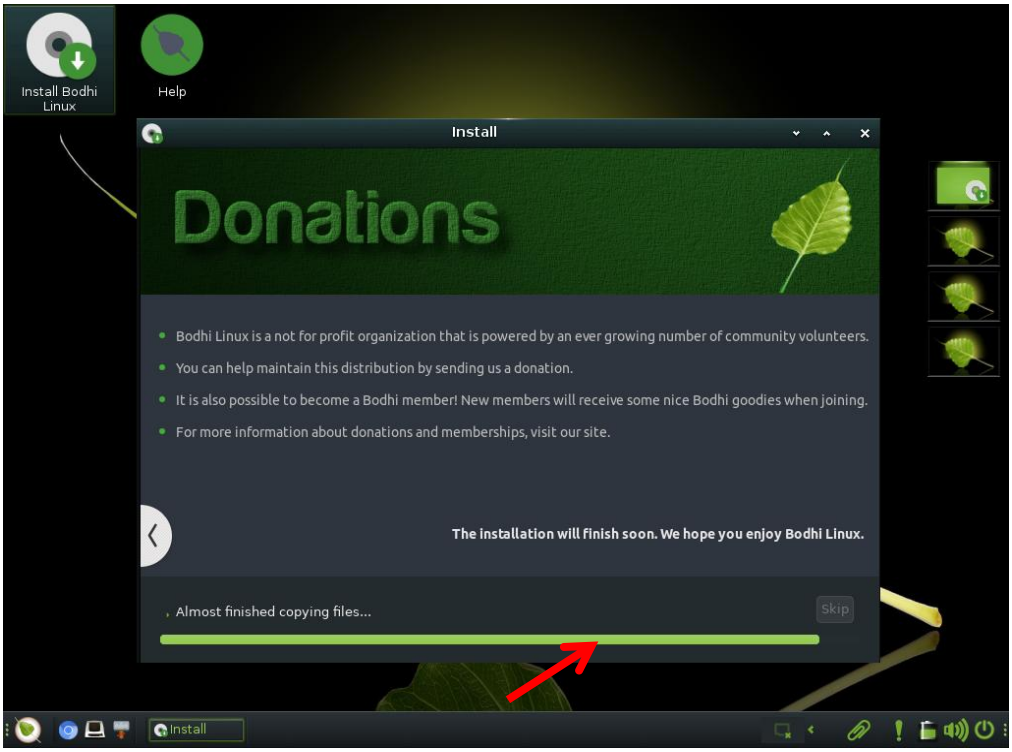

Fig 3.11 System File Copying

Step 12: Installation Box Dialogue Box

Fig 3.12 Installation Complete Dialogue

Step 13: Installation Medium Home Page

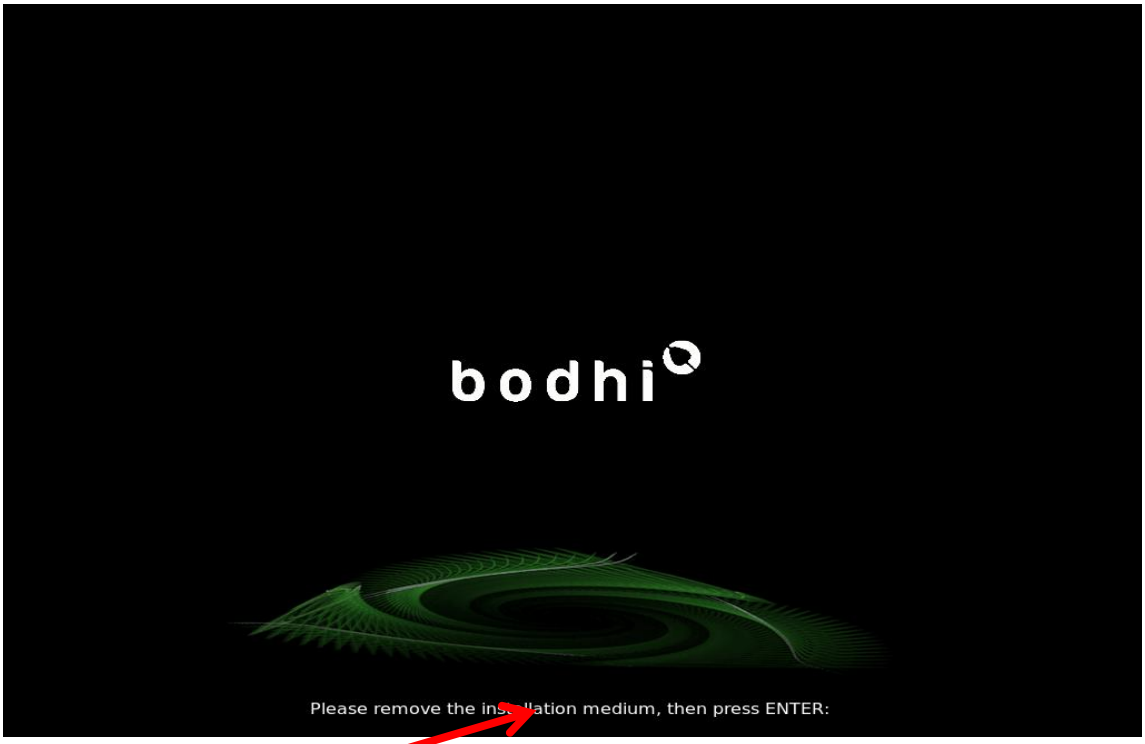

Fig 3.13 Removing installation medium home page after restart

Step14: Removing Disk from Virtual Box

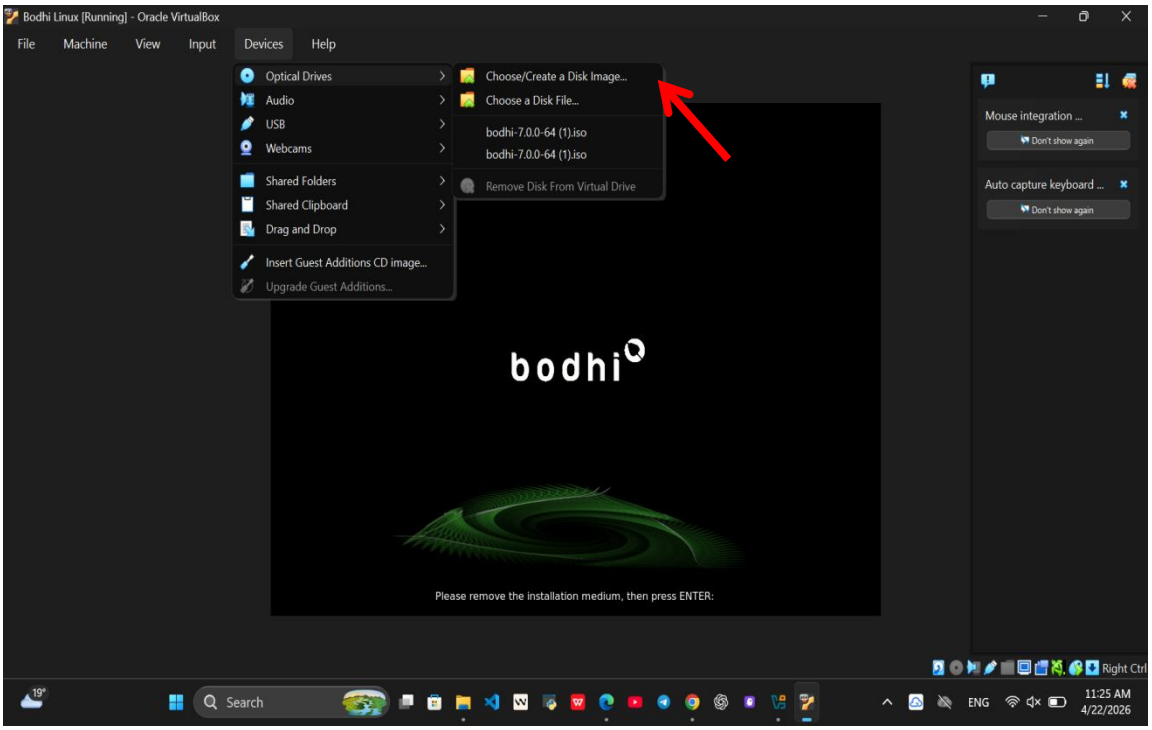

Fig 3.14 RemovingDisk from Virtual box

Step 15: Final Installed Desktop Verification Home Page of Bodhi Linux

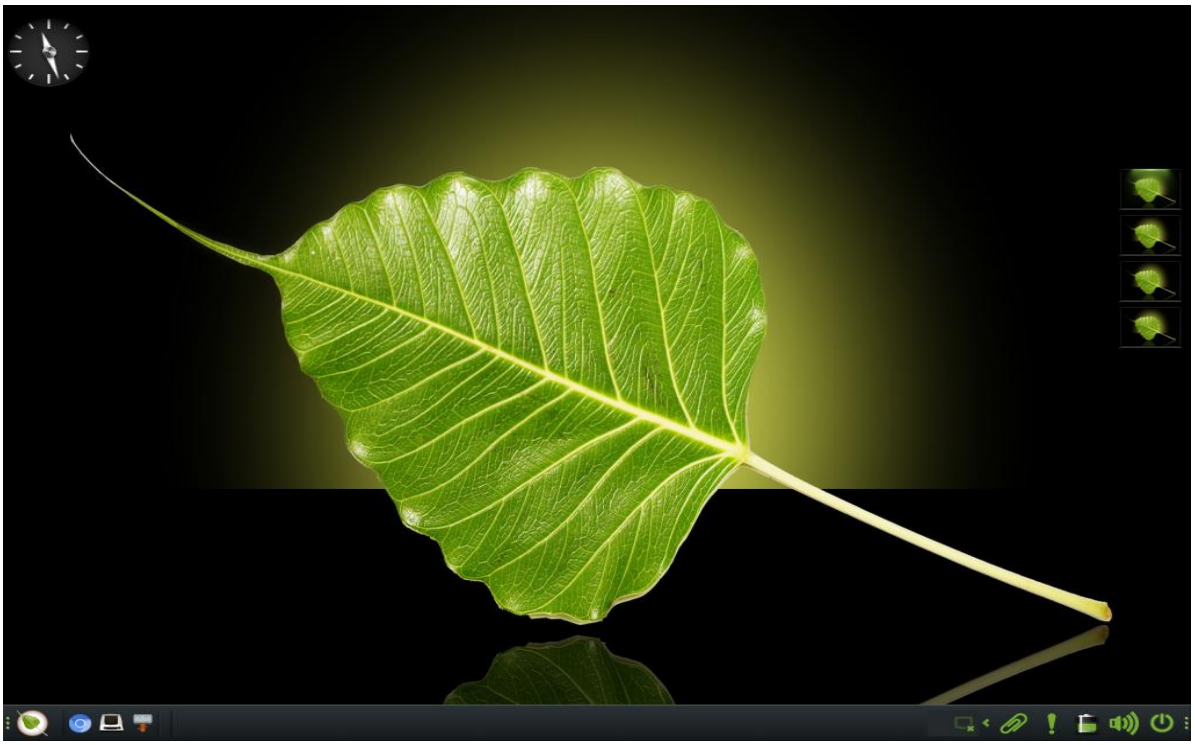

Fig 3.15 HOME PAGE OF Bodhi-Linux

Step 16: Requiring User Password to LOGIN

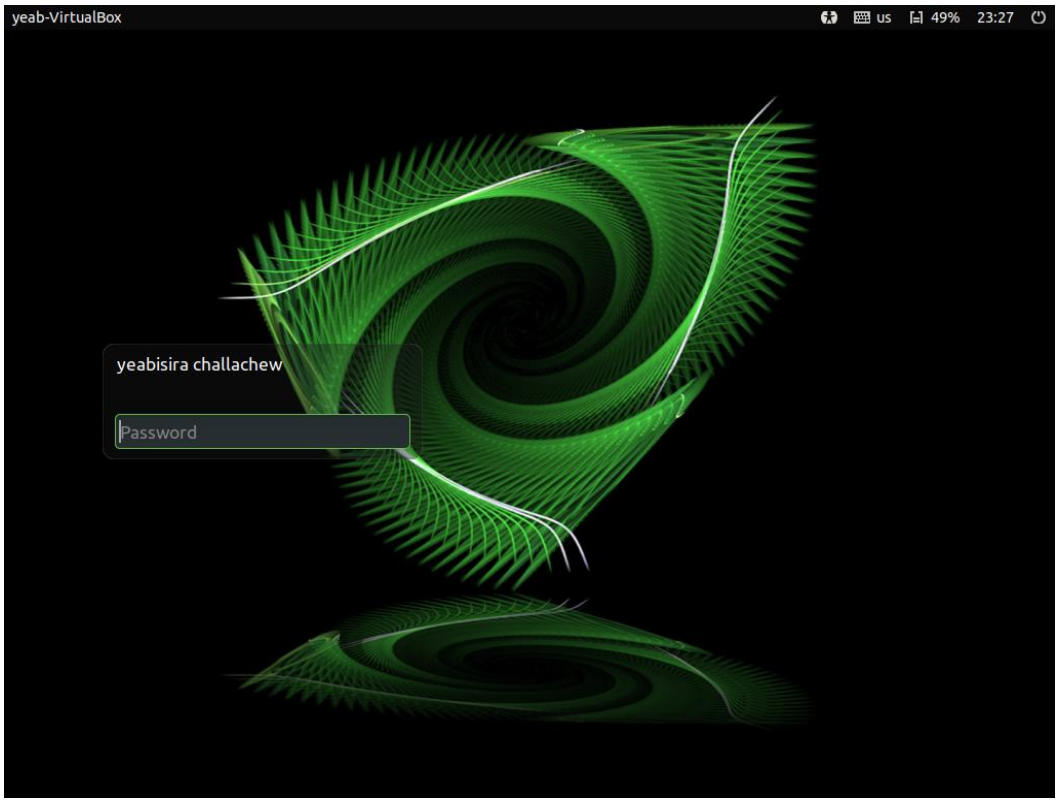

Fig 3.16 Final Installed Desktop Verification home page

Step 17: Checking the Username in Terminal/ Command Prompt

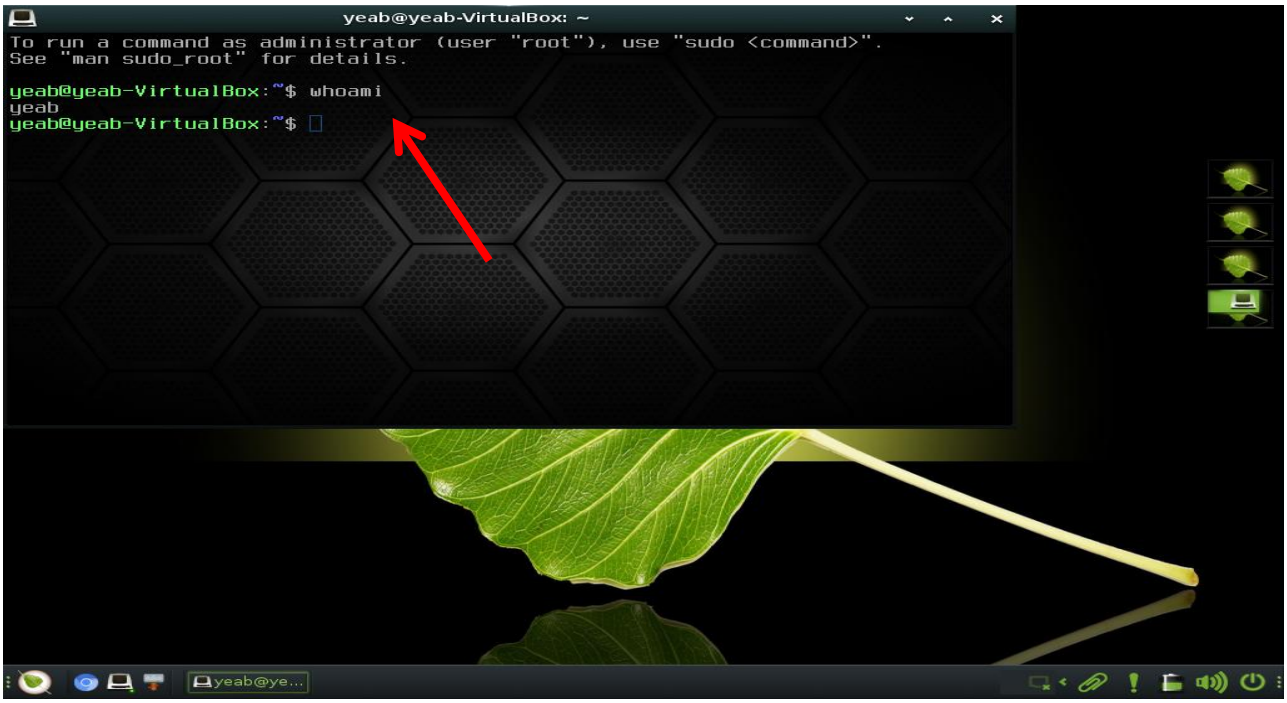

Fig 3.17 Checking the NAME of the User Account using CMD PROMPT

D) Issues(Problem Faced) & Solutions

1. - Issue : I faced a connection issue while accessing the official website

- **Solution :** I used a mirror link from Source-Forge to download the ISO

2. - Boot Order Issue: After installation, the system attempted to boot from the ISO again.

- **Solution:** Manually removed the installation medium via "**Devices > Optical Drives > Remove disk from virtual drive**."

3. - Issue: Application Execution Error- Install Bodhi Linux stopped running unexpectedly

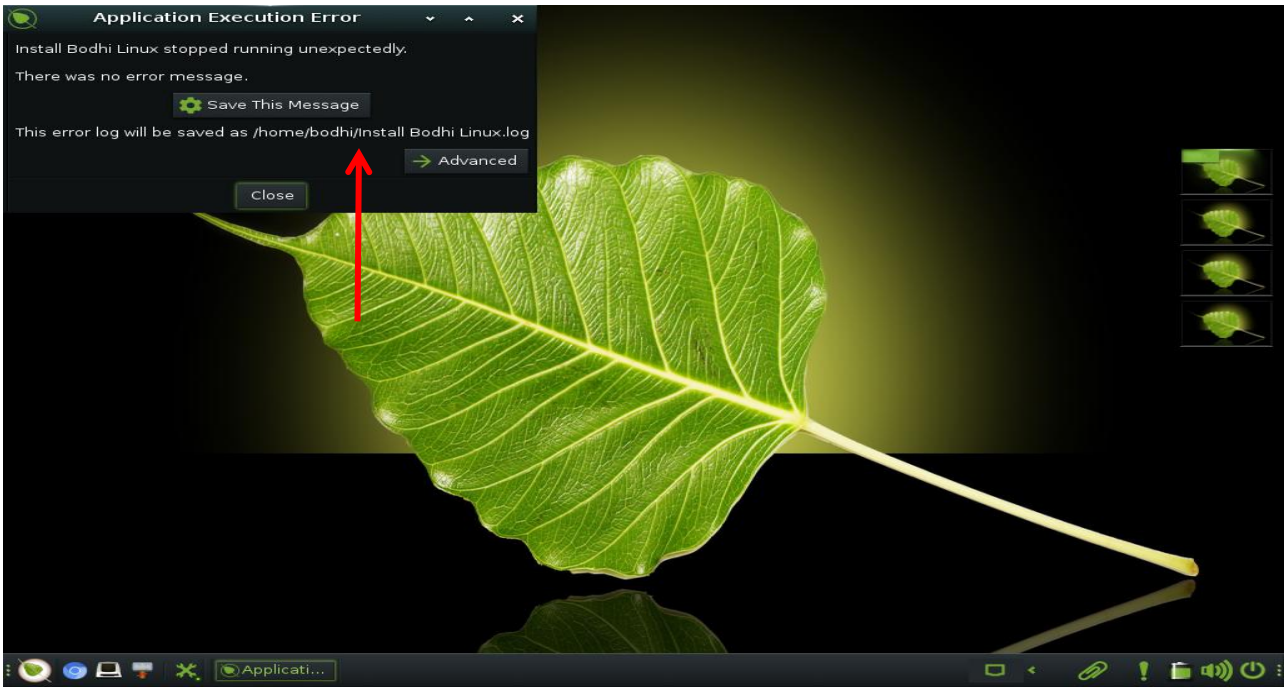

Fig 4.1 Application Execution Error

– Solution:

I restarted the installer and increased the virtual RAM to 4GB, which resolved the crash.

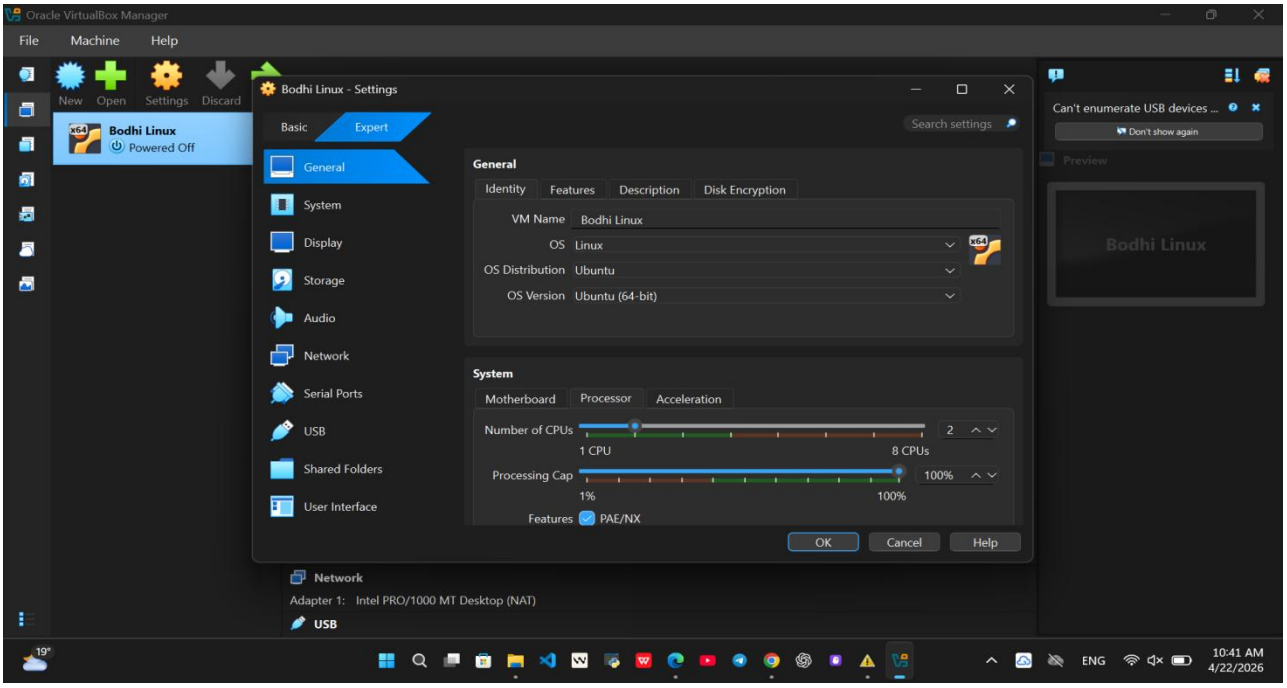

Fig 4.2 PAE/NX Enabling

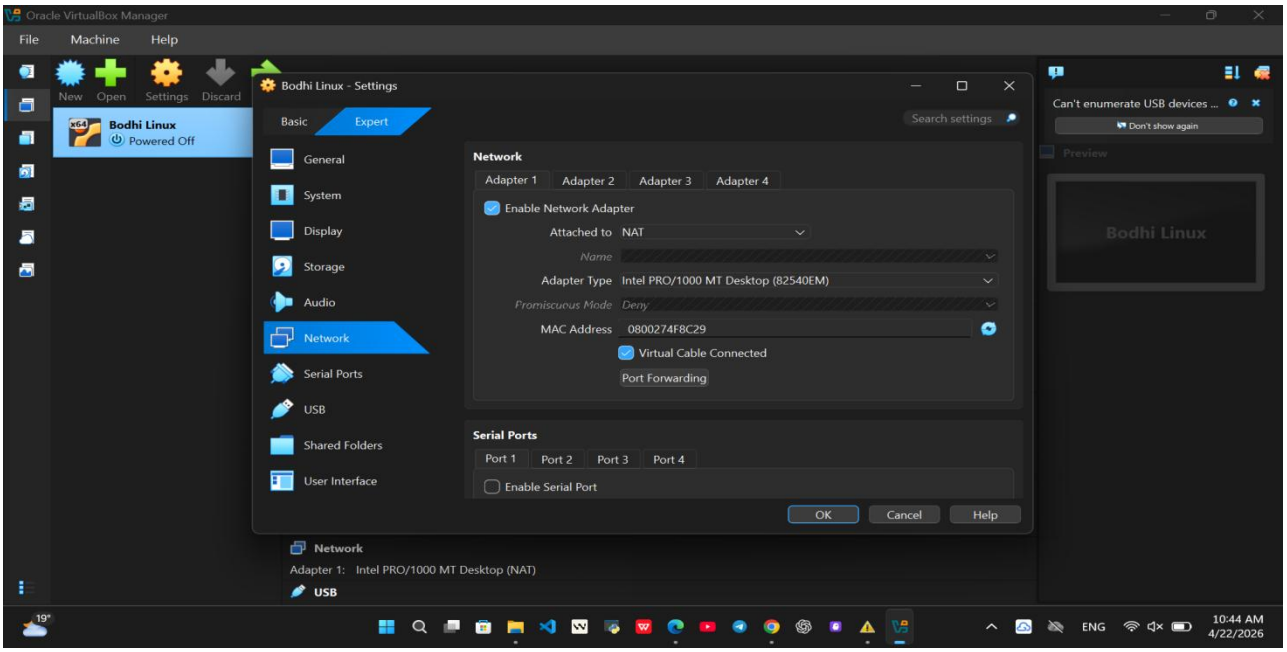

Fig 4.3 Home Page OF NETWORK Connected

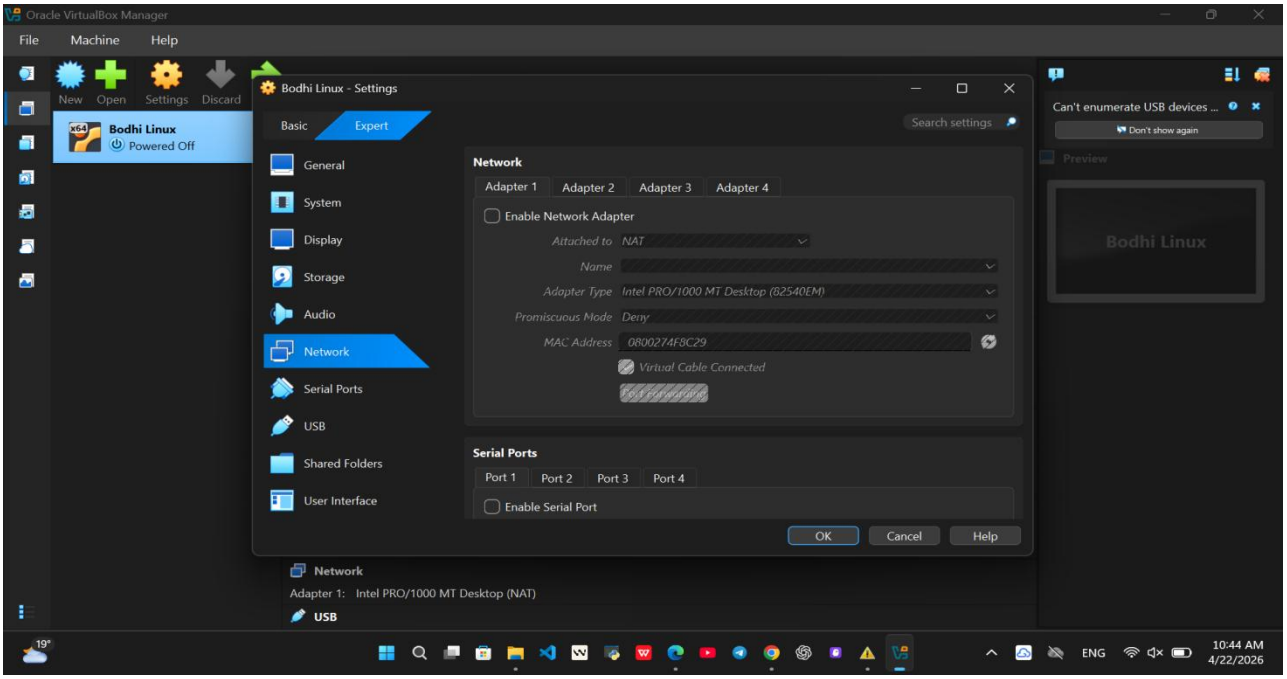

Fig 4.4 disabled the Network Adapter

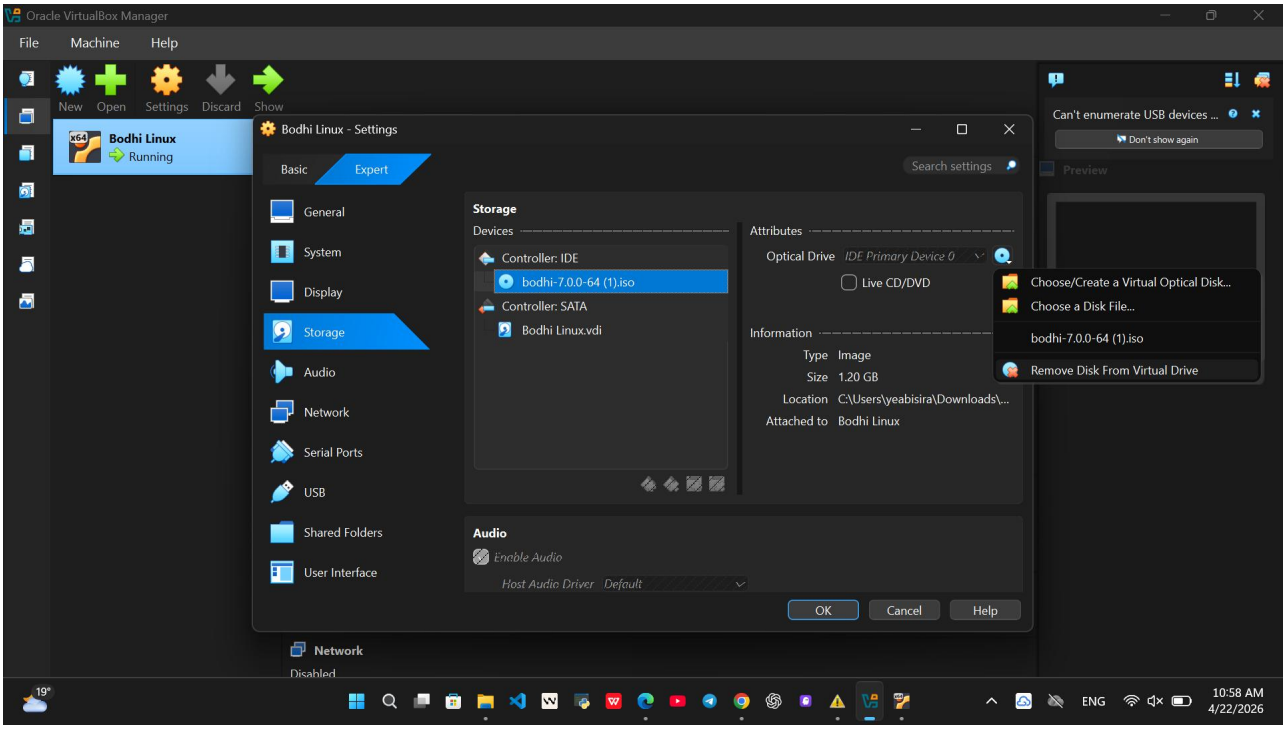

Fig 4.5 Home Page for Removing Disk from Virtual Drive

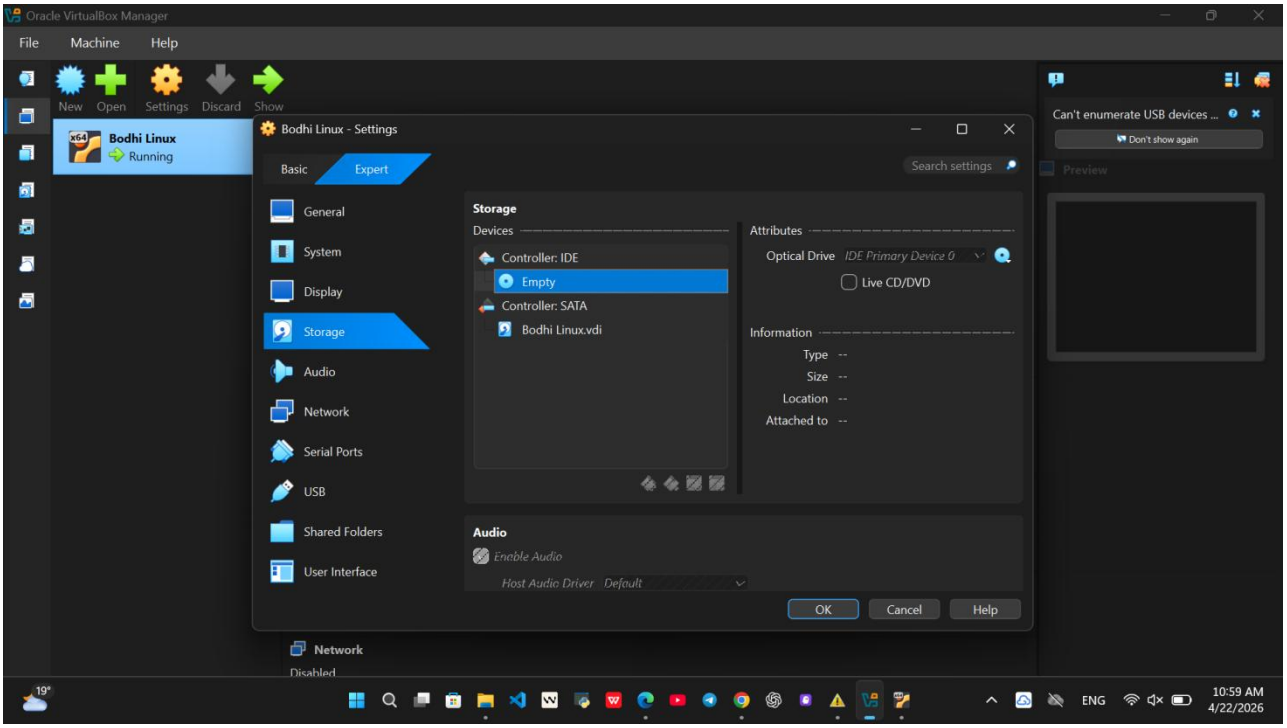

Fig 4.6 After removing the file

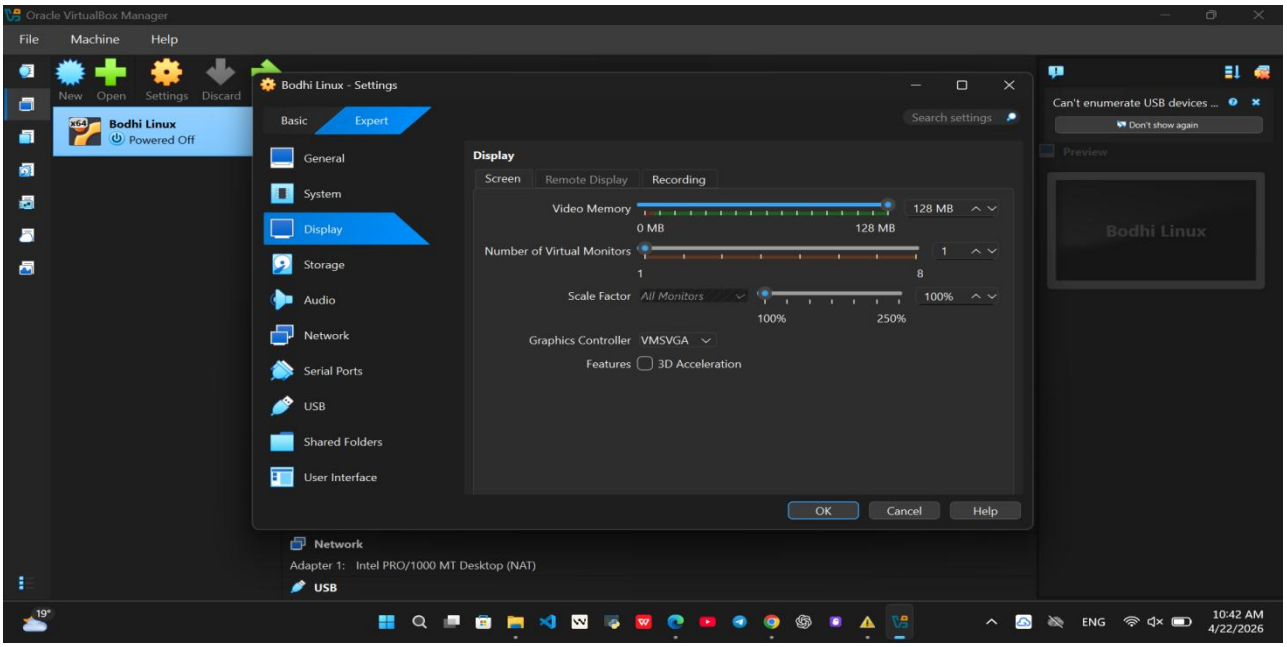

Fig 4.7 increasing video memory

F) Filesystem Support

- Bodhi Linux primarily supports and uses **ext4 (Fourth Extended Filesystem)**.

Why: ext4 is the standard for Linux because it supports huge file sizes and volumes, uses journaling to prevent data corruption during power failures, and offers faster disk checking compared to older versions. It is highly reliable for a Lightweight OS like Bodhi.

G) Advantage and Disadvantage

Advantages (Strengths):

- **Extremely Lightweight:** Bodhi Linux uses very few system resources (RAM and CPU), making it one of the best choices for reviving older computers or running on low-end hardware.
- **Moksha Desktop Environment:** The Moksha desktop is fast, efficient, and highly customizable. It allows users to change every aspect of the desktop's appearance and behavior through various themes and gadgets.
- **Ubuntu-Based Stability:** Since it is built on the Ubuntu Long Term Support (LTS) core, it is very stable, secure, and has access to a massive repository of software and security updates.
- **Minimalist Philosophy:** It comes with only essential software pre-installed (like a web browser and a file manager). This prevents "bloatware" and gives the user full control over what applications they want to install.

Disadvantages (Weaknesses):

- **Learning Curve for Beginners:** While it is based on Ubuntu, the Moksha desktop operates differently than Windows or macOS. New users might find it a bit confusing at first and may need to use the terminal for certain configurations.
- **Minimal Out-of-the-Box Software:** Because it is minimalist, users must manually install basic tools like Office suites (LibreOffice), media players, or image editors, which might be inconvenient for some.
- **Smaller User Community:** Compared to giant distributions like Ubuntu, Linux Mint, or Fedora, Bodhi has a smaller community. This means finding specific tutorials or community-driven support can sometimes take longer.
- **Aesthetic Appeal:** The default look of Bodhi can feel "old-fashioned" to some users compared to modern, flashy desktop environments like GNOME or KDE Plasma.

Comparison Table

Advantages	Disadvantages
Runs fast on old or low-spec hardware.	Can be challenging for absolute beginners.
Highly customizable and flexible UI.	Requires manual installation of most apps.
Stable and secure (Ubuntu-based).	Smaller community for support.
Minimalist (No unnecessary bloatware).	Default design may feel outdated to some.

H) Conclusion

The installation of **Bodhi Linux** on **Oracle VM VirtualBox** was completed successfully. This project demonstrated the efficiency of lightweight Linux distributions in a virtualized environment. Although technical hurdles were encountered—specifically the installer crash and the post-installation boot error—they provided a valuable opportunity to apply troubleshooting techniques, such as adjusting hardware resource allocation (Video Memory) and managing virtual optical drives.

The final system is fully functional, with the user account successfully configured under my full name (**Yeabisira Challachew**) and the **ext4** filesystem properly initialized. This project confirms that virtualization is a powerful tool for testing and deploying diverse operating systems safely and efficiently without altering the host hardware.

I) Future Outlook / Recommendation

Future Outlook:

As hardware continues to evolve, the demand for lightweight and resource-efficient operating systems like Bodhi Linux will remain high, especially for IoT (Internet of Things) devices and the repurposing of older computer hardware. Virtualization technology will continue to be the standard for cloud computing and secure software testing.

Recommendations:

> **For Beginners:** I recommend starting with at least **4GB of RAM** and **128MB of Video Memory** in VirtualBox settings before starting the installation to avoid the common crashes I experienced.

For Performance: Users should install the **"VirtualBox Guest Additions"** after the OS is installed to enable better screen resolution and smoother mouse integration.

For Security: I recommend always choosing a distribution based on **LTS (Long Term Support)** like Bodhi to ensure the system receives security patches for several years.

What is Virtualization?

Virtualization is a technology that allows you to create multiple simulated environments or dedicated resources from a single, physical hardware system. It uses a thin layer of software called a **Hypervisor** (such as Oracle VM VirtualBox) to separate the physical hardware from the operating systems. In simple terms, it lets you run a "computer within a computer."

Why Virtualization?

Virtualization has become a standard in modern computing for several critical reasons:

- **Resource Efficiency:** It allows multiple operating systems (like Windows and Linux) to run on one physical machine, ensuring that hardware power is not wasted.
- **Cost Savings:** It reduces the need to buy multiple physical computers for different tasks, saving money on hardware, electricity, and space.
- **Isolation and Security:** Each Virtual Machine (VM) is isolated. If one VM crashes or gets infected with a virus, it does not affect the host computer or other VMs.
- **Software Testing:** Developers can test new software on different operating systems safely without risking their main system.

How Virtualization Works?

Virtualization works through a software layer known as the **Hypervisor**. There are two main ways it functions:

- **The Hypervisor Layer:** The hypervisor sits between the physical hardware and the virtual machines. It intercepts commands from the Guest OS (e.g., Bodhi Linux) and translates them so the physical hardware can understand them.
- **Resource Allocation:** When you create a VM, the hypervisor "borrows" a specific amount of RAM, CPU power, and Disk space from the physical (Host) machine and gives it to the Guest OS.
- **Encapsulation:** The entire virtual computer is saved as a few files on your hard drive (like the .VDI file we created). This makes it easy to move, copy, or delete the entire operating system just like a regular file.

THANK

YOU !!!